# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER ARCHITECTURE COURSE CODE: CSA12**

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**COURSE OUTCOME COVERED**

**CO1**: Components of a computer system, including CPU, memory, and I/O. Basic Operational Concepts: Instruction execution cycle, instruction fetch and decode.Bus Structures: Single-bus, multi-bus, and hierarchical buses.Factors of Performance: CPU execution time, CPI, clock cycles, and benchmarking.8085 and 8086 Architectures: Overview, instruction sets, and addressing modes. Modern Architectures: ARM Cortex, Intel Core, and AMD Ryzen.Instruction Sets, Formats, and Addressing Modes. Number Systems: Binary, hexadecimal, and their applications in computer systems. (BL4)
Assessment Tool Weightage: 40%

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**QUESTIONS:5** **Total Marks:25**

**ANNEXURE A**
**CONCEPT MAPPING**

***Code of Conduct:***

*I Brasilla R (192511357)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

## ANNEXURE A

1. Construct a concept map that connects the components of a computer system (CPU, memory, and I/O) with the instruction execution cycle. Clearly show how instruction fetch, decode, and execution stages interact with memory and input/output operations during program execution.
2. Develop a concept map linking different bus structures (single-bus, multi-bus, hierarchical bus) with system performance factors such as CPU execution time, clock cycles, and CPI. Illustrate how bus design influences data transfer speed and overall system efficiency.
3. Create a concept map that compares 8085 and 8086 architectures with modern architectures like ARM Cortex, Intel Core, and AMD Ryzen. Show relationships in terms of instruction sets, addressing modes, processing capability, and architectural advancements.
4. Design a concept map that integrates instruction sets, instruction formats, and addressing modes with number systems (binary and hexadecimal). Explain how instructions are represented, decoded, and executed using these number systems in a computer.
5. Prepare a concept map that connects benchmarking techniques with performance metrics such as CPU execution time, CPI, and clock cycles, and relate these to different architectures (8085, 8086, and modern processors). Demonstrate how performance evaluation varies across different system designs.

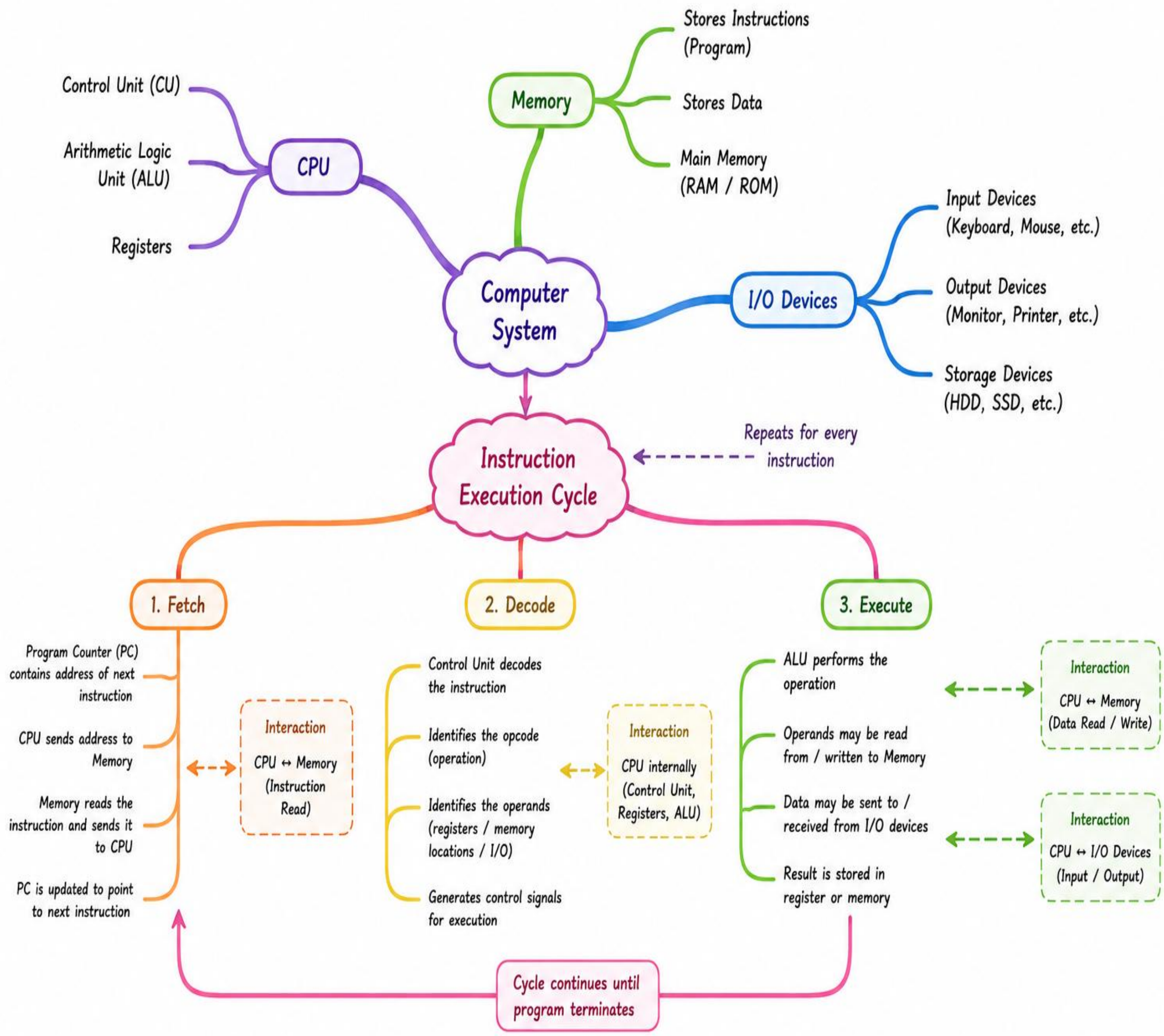
**MARKS ALLOCATION**

| Criteria | Understanding of Problem Context (20%) | Application of Concepts (20%) | Problem-Solving Approach (20%) | Accuracy of Solution (20%) | Clarity (20%) |
|---|---|---|---|---|---|
| Q1 | | | | | |
| Q2 | | | | | |
| Q3 | | | | | |
| Q4 | | | | | |
| Q5 | | | | | |

## RUBRICS FOR CALCULATION:

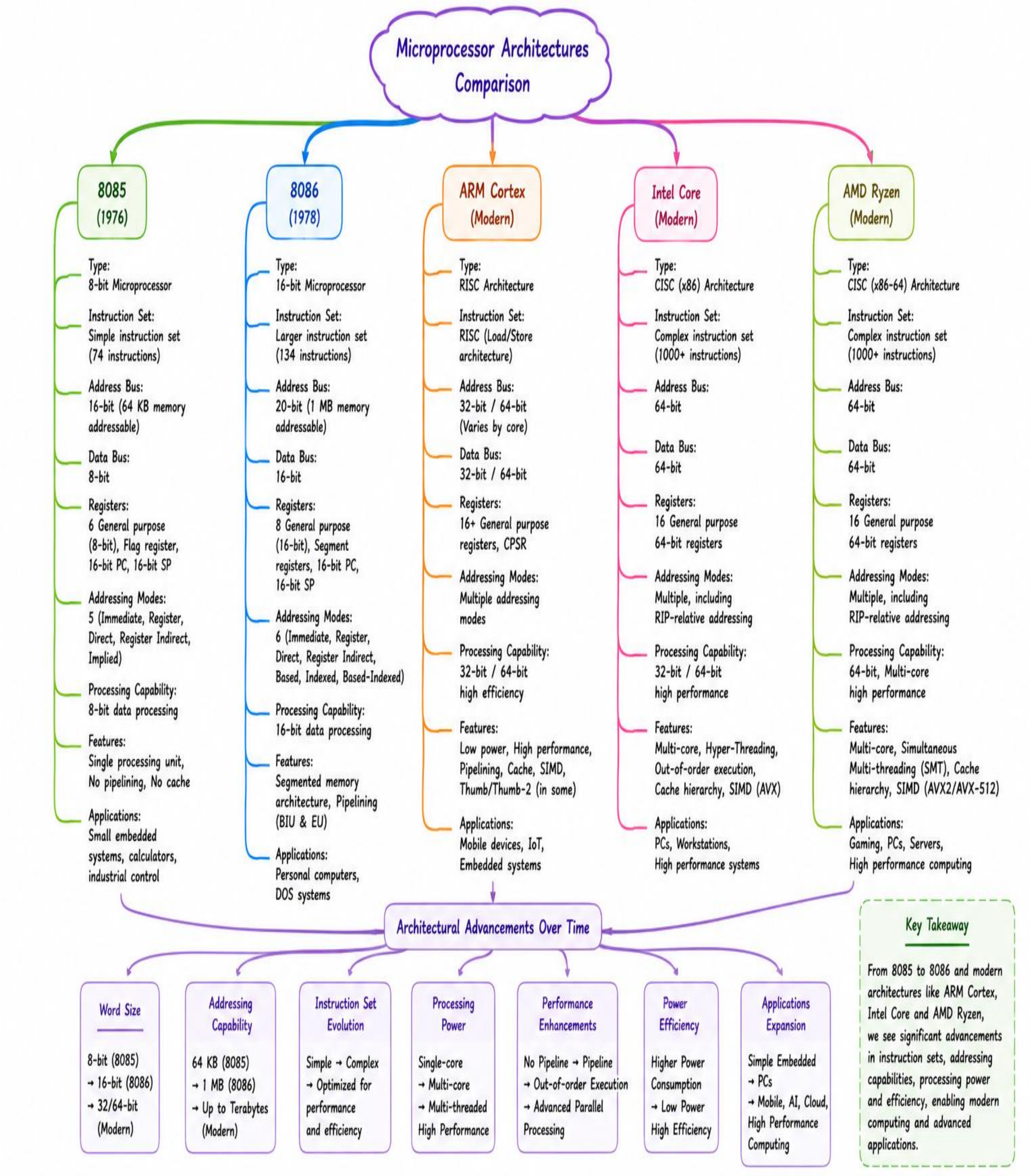
| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Problem Context | 20 | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts | 30 | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy | Fails to apply appropriate concepts |
| Problem-Solving Approach | 20 | Logical, well-structured, and efficient approach | Mostly logical approach with minor gaps | Basic approach with some inconsistencies | Illogical or unclear approach |
| Accuracy of Solution | 20 | Completely correct solution with proper justification | Mostly correct with minor errors | Partially correct solution | Incorrect or incomplete solution |
| Clarity | 10 | Clear, well-organized, and properly explained solution | Understandable with minor clarity issues | Limited clarity and explanation | Poorly presented and difficult to understand |

# 1) Execution Cycle



## 2) Bus structure and system performance

3) Microprocessor architecture comparison



4)Addressing modes and number systems

5)Performance metrics and architecture comparison